

SPI-M Consensus Statement on: Epidemiology of the A (H1N1) Strain and its Implications for the Advised Planning Assumptions

18th June 2009 - Final

Epidemiology of the A (H1N1) strain outside of the UK.

At this time, uncertainty remains about all aspects of the outbreak, and so any parameter estimates or judgements must be regarded as preliminary, uncertain and liable to change as further data become available. Care needs to be taken when considering parameters estimated from non-UK data. However, data from the initial foci of the epidemic (in particular the early data from Mexico and the epidemics in North America) gives an indication of the epidemiology that might be expected.

Data from well-documented outbreaks that have gone to completion allows the clinical, community attack rate (CAR) to be estimated. One outbreak in the village of La Gloria, Mexico gives an overall estimate of 39%¹ with a significantly higher CAR in children (aged 15 or less) compared to adults. However, only one of these cases was confirmed A(H1N1) influenza and A(H3N2) and Influenza B infections were also present.

Households provide a well-defined exposure setting in which to measure attack rates. Currently estimates from the United States, where it is not policy to give household contacts prophylaxis, suggest that household attack rates are in the region 10-20%. These are towards the lower part of the range for seasonal influenza (typically 10%-40%). In general, household attack rates would be expected to be higher than community attack rates.

School outbreaks also provide a well-defined setting to assess clinical attack rates. In one well-documented and large outbreak in a boarding school in New York, a total of 662 cases of H1N1 were observed over a period of 1 month, of which 642 were in pupils. This gives an attack rate of 24% in pupils and 9% in staff (significantly higher in children). It should be noted that higher attack rates for seasonal influenza have generally been observed in boarding schools.

Taken together these various sources of evidence suggest population attack rates for the UK could be expected to be in the range 20 to 30% but up to twice as high in children.

The case fatality ratio is uncertain and is likely to remain so for two reasons. First, not all probable cases are now being confirmed in the laboratory, those with mild symptoms may never contact health services, and hence the total number of cases will be under-ascertained. Secondly, there can be a delay of up to 2-3 weeks between onset of symptoms and death for severe cases and so for example, the ratio of reported deaths to confirmed cases in the US may rise due to this delay. However, the outbreak in the village of La Gloria, Mexico, along with estimates considering Mexico as a whole, suggest the CFR lies towards the lower end of a 0.2% - 1.4%

¹ Christophe Fraser et al, Scienceexpress 11th May 2009, Pandemic Potential of a Strain of Influenza A(H1N1): Early findings.

range. These estimates are, again, broadly consistent with those based on data from elsewhere.

Estimates of the reproduction number, R , from Mexico vary from 1.2 – 1.6¹. Confidence intervals of these estimates are wider than this, though they are narrowing. Estimates of the serial interval in Mexico are 1.3 – 2.7 days¹. Taken together, this suggests a doubling time of around 7 days. These estimates are also generally consistent with those based on data from the US and Canada.

Estimates of the hospitalisation rates vary from around 2% to 10%. This wide range is likely to reflect different criteria for hospitalisation in different countries as well as under ascertainment of mild cases. About half of these hospitalisations are in the clinical seasonal influenza risk groups and pregnant women (a similar pattern is seen for deaths).

The epidemic continues to spread globally and the number of countries affected is increasing steadily. As the size of the epidemic increases in individual countries however, there has been a tendency for the detail of reporting to decrease. Hence, later data tend to be less informative than those from early in the first outbreaks. While such data is a useful test of the estimates from the first foci, and is, as noted above, consistent with those estimates, it is unlikely in general², to lead to any substantial further refinement of these parameter estimates, based purely on data from abroad. Good UK data is therefore going to be critical to improving model parameterisation and our future forecasts.

Epidemiology of the A (H1N1) strain inside the UK.

All results based on analyses of UK data are uncertain and preliminary. In particular, parameter estimates are based on epidemiological linkages between cases, the investigation of which is still underway in many instances. The results are therefore liable to change, reflecting the stochastic nature of the processes and the unfolding epidemiological picture.

The serial interval (the interval from symptom onset in primary cases to onset in their secondary cases) can be estimated directly from linked pairs of cases. The modal serial interval appears to be 3 days, with a range of between 0 and 10 days. It is not possible to make robust estimates of incubation periods and infectious periods from current data.

Estimates of the reproduction number, R , can be made using a variety of methods, including analysis of the cluster size distribution, and infection trees. The consensus estimates are that the reproduction number for novel H1N1 in the UK is currently around 1, with confidence intervals generally bracketing 1. Currently best estimates for children are higher than those for adults. As cases tend to arise from clusters, estimates tend to vary from day to day, however there is a general upward trend in estimates over time. The increase in R may represent factors such as the disease moving into different groups and longer periods before HPA interventions become effective in controlling the clusters.

² An exception would be high quality data from one or more southern hemisphere countries.

The outbreak is currently concentrated on children and young adults. This is consistent with the behaviour of previous pandemics, in particular 1957, and seasonal influenza where initial spread is in children and attack rates tend to peak in school age children. This is also consistent with a higher level of susceptibility in children. Estimates suggest that children are at least twice as susceptible as adults. Early serological analyses seems to suggest significant immune responses in the elderly, some in younger adults but little in children³.

Most cases that are picked up by the surveillance systems receive antivirals. Where antivirals have been given promptly (within 3 days of symptoms) it has been estimated that there has been a 20% drop in transmission in line with pre-outbreak expectations. Given that there are often greater delays the overall effect of the current policy is probably around a 10% decrease in transmission.

Based on the cases in the UK to date an upper bound on the case fatality ratio is estimated to be 0.9%. This estimate has not been affected by the recent fatality.

Hospitalisation rates are estimated in the range 1.5% to 3 %.

Levels of novel H1N1 influenza illness remain small compared with expected background levels of ILI, and even assuming significant growth of the current outbreak, would be expected to do so for some weeks.

The timing of the first (or only) major wave of infection is uncertain due to the impact of imports of cases from abroad and seasonal effects. If the current growth in cases is sustained, a substantial wave of clinical cases with an attack rate say of around 25% could peak in late August. Alternatively, smaller wave of infections of possibly half this size could peak earlier in August. Equally plausible scenarios suggest no major wave of infection until January 2010.

³ Kat et. Al. <http://www.cdc.gov/mmwr/preview/mmwrhtml/mm5819a1.htm>

Advised Planning Assumptions

SPI-M considers there is sufficient evidence to *modify the standard reasonable worst case planning assumptions* as follows:

- For the current outbreak (that is until the end of the first major wave of infection), up to 30% clinical attack rate (CAR) averaged across the population (with serological rates up to 60%). The clinical attack rates in children may be up to twice that in adults (up to around 50%).
- Peak illness rates of around 7% (measured in new cases per week as a proportion of the population) in the peak fortnight.
- Absences rates for illness reach 10 - 12% in the peak weeks (assuming an average 7 working day absence for those without complications, 10 for those with and some allowance for those at home caring for children).

And in particular for the current outbreak over the spring/summer:

- 0.9% 'reasonable worst case' case fatality ratio (CFR),
- Complication rates up to 15%,
- Case hospitalisation demand rates of up to 3%, of which 25% could, if the capacity existed, require intensive care.

However, planning for possible later development of the current pandemic, such as a 'second wave' of a substantially mutated virus, as for any future outbreak should be based on the standard reasonable worst case assumptions.

For reference, the *standard planning assumptions* based on a 'reasonable worst case' are⁴:

- Up to 50% cumulative clinical attack rate (CAR) (with serological rates up to 80-85%).
- Of which, from 10% up to 25% are expected to have complications, half of these bacteriological.
- Peak illness rates of around 10 - 12% (measured in new cases per week as a proportion of the population) in the peak fortnight.
- Absences rates for illness reach 15-20% in the peak weeks (at a 50% overall attack rate, assuming an average 7 working day absence for those without complications, 10 for those with, and some allowance for those at home caring for children).
- Case hospitalisation demand rates in the range 0.55% to 4% with an average six day length of stay, of which 25% could, if the capacity existed, require intensive care for 10 days.
- Case fatality rates in the range 0.4% to 2.5%.

These advised planning assumptions will be kept under review, and are liable to change as further data become available on the current strain of Influenza A (H1N1)

⁴ September 2008: Scientific Pandemic Influenza Advisory Committee (SPI): subgroup on Modelling available at <http://www.advisorybodies.doh.gov.uk/spi/modelling.htm>